

1. Administrative & Study Identification

Full Study Title: A randomized, double-blind, placebo controlled, 2-arm multicenter phase 3 study to assess the efficacy and safety of ianalumab in patients with active Sjogren's syndrome **Short Title/Acronym:** NEPTUNUS-1 **Protocol Number:** CVAY736A2301 **NCT Number:** NCT05350072 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Ianalumab (VAY736) **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To demonstrate the clinical efficacy, safety, and tolerability of ianalumab (VAY736) administered subcutaneously (s.c.) monthly compared to placebo in reducing systemic disease activity in patients with active Sjögren's syndrome. **Secondary Objectives:** To evaluate patient and physician global assessments, changes in symptom evaluations (such as dryness and fatigue), and the long-term safety profile over 52 weeks.

2.2 Background & Rationale To address the unmet medical need for patients with active Sjögren's disease, a progressive, systemic autoimmune disorder that currently lacks approved targeted therapies. Ianalumab targets the underlying B-cell-driven autoimmune response through a dual mechanism of action: B-cell depletion via antibody-dependent cellular toxicity (ADCC) and inhibition of B-cell activating factor receptor (BAFF-R) signaling. This study aims to provide robust Phase III evidence that ianalumab significantly reduces disease activity and patient symptom burden.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (1:1, 2-arm)

3.2 Planned Enrollment & Duration Target \$N\$: 275 adult patients **Number of Sites:** Global, Multicenter **Estimated Study Start:** May 2022 **Estimated Primary Completion:** Late 2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Women and men ≥ 18 years of age. 2. Classification of Sjögren's syndrome according to the ACR/EULAR 2016 criteria with a time since diagnosis of ≤ 7.5 years at screening. 3. Positive anti-Ro/SSA antibody at screening (patients negative for anti-Ro/SSA are eligible if they have a positive salivary gland biopsy confirmed by central expert review). 4. Screening ESSDAI (EULAR Sjögren's Syndrome Disease Activity Index) score of ≥ 5 within 8 specified domains.

4.2 Exclusion Criteria 1. Presence of another autoimmune rheumatic disease that is active and constitutes the principal illness. 2. Prior treatment with ivalumab or use of other B-cell depleting therapies within 36 weeks prior to randomization. 3. Active viral, bacterial, or other infections requiring systemic treatment (e.g., active tuberculosis, active Hepatitis B or C). 4. Pregnant or nursing (lactating) women.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ivalumab 300 mg administered once monthly (every 4 weeks). **Route of Administration:** Subcutaneous (s.c.) injection. **Duration of Treatment:** 52 weeks (with an option to enter a long-term extension trial).

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of hydroxychloroquine (≤ 400 mg/day), methotrexate (≤ 25 mg/week), azathioprine (≤ 150 mg/day), or systemic corticosteroids (≤ 10 mg/day prednisone equivalent) for at least 30 days prior to randomization. **Prohibited:** Other investigational drugs, unapproved DMARDs, and specified Traditional Chinese Medicines (unless appropriate washout periods are met).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Up to -28 Days Informed Consent, Medical History, Labs, ACR/EULAR 2016 verification, ESSDAI Baseline
Baseline	Day 0	Randomization, First Subcutaneous Dose, PRO Diary Initialization	Interim
Safety Labs, Efficacy Assessment (ESSDAI), Symptom Diaries, IP Administration	End of Study	Week 48 / 52	Final ESSDAI assessment, Exit Interview, Transition to Long-Term Extension / Safety Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change from baseline in systemic disease activity at Week 48. **Measurement Method:** EULAR Sjögren's Syndrome Disease Activity Index (ESSDAI) clinical assessment scale.

7.2 Secondary & Exploratory Endpoints * Patient and physician global assessments of disease activity. * Symptom evaluations including dryness and fatigue via the Sjögren's Syndrome Symptom Diary (SSSD). * Safety and tolerability profile (AEs and SAEs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard continuous collection via patient diaries, scheduled onsite interviews, and laboratory monitoring (with specific tracking for B-cell count recovery and viral markers like HBV/HCV). **Serious Adverse Events (SAEs):** Standard reporting timeline, typically within 24 hours of site awareness. **Data Monitoring Committee (DMC):** External independent committee to review unblinded safety and efficacy data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy evaluation; Safety Set for all participants receiving at least one dose. **Sample Size Justification:** Target $N=275$ powered ($\alpha = 0.05$, $\text{Power} \geq 80\%$) to detect a statistically significant and clinically meaningful reduction (negative change) in ESSDAI scores compared to the placebo arm. **Handling of Missing Data:** Mixed-Model for Repeated Measures (MMRM) and/or Multiple Imputation for continuous efficacy endpoints such as ESSDAI.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite and remote source data verification by sponsor clinical research associates (CRAs).

Audit Trail: Robust electronic Data Capture (eDC) system with eCRF locking, automated data clarification form (DCF) generation, and centralized biopsy read verifications.

11. Study Identifiers & Governance

Primary ID: CVAY736A2301 **Registry Number:** NCT05350072 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** NEPTUNUS-1 **Official Regulatory Title:** Two-arm Study to Assess

12. Clinical Framework (Sjögren's Disease Specific)

Primary Condition: Active Sjögren's Syndrome (Sjögren's Disease)
Diagnosis Threshold: ACR/EULAR 2016 classification criteria & ESSDAI score ≥ 5
Medical Methodology: Monoclonal antibody targeting B-cells (B-cell depletion & BAFF-R blockade)
Optimization Target: Clinically meaningful reduction in ESSDAI score and patient symptom burden (dryness/fatigue)

13. Study Procedures & Methodology

Management Pathway: Subcutaneous biologic therapy administration
Education Protocol (HCP): Subcutaneous injection protocols, AE monitoring for immunosuppression, and standardized ESSDAI scoring
Education Protocol (Patient): Self-reporting symptom tracking (SSSD) and monitoring for signs of infection
Quality Audit Mechanism: Central expert review of salivary gland biopsies (for anti-Ro/SSA negative patients) and automated eDiary adherence tracking

14. Observation & Follow-Up Schedule

Total Duration: 52 weeks (plus optional long-term extension or safety follow-up until B-cell recovery)
Onsite Visit Cadence: Monthly during the subcutaneous dosing cycle (with major clinical assessments at Weeks 16, 24, 36, 48, and 52)
Remote Monitoring Frequency: Continuous electronic Patient-Reported Outcomes (ePRO)
Checklist Review Interval: Every 4 weeks (Q4W) prior to subsequent IP dosing

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				